

Optimal Operational Plan: Instance 05

Total Summary

This plan outlines the optimal vehicle dispatching strategy. By successfully fulfilling 8 out of 10 orders (80% fulfillment rate), the system maximizes resource utilization and generates an **expected total profit of \$106,800**, while strictly adhering to relocation budget and time constraints.

1. Order Fulfillment Strategy

Action Items:

- **Cancellations:** Notify clients of **Orders 01 and 02** regarding their cancellations due to capacity optimization.
- **Upgrades:** Inform the client of **Order 06** about their complimentary upgrade to a Level 2 vehicle.

Order ID	Status	Assigned Vehicle	Requested Level	Assigned Level	Free Upgrade
Order 01	Rejected	-	L1	-	-
Order 02	Rejected	-	L1	-	-
Order 03	Accepted	Vehicle 03	L1	L1	No
Order 04	Accepted	Vehicle 01	L1	L1	No
Order 05	Accepted	Vehicle 06	L2	L2	No
Order 06	Accepted	Vehicle 04	L1	L2	Yes
Order 07	Accepted	Vehicle 05	L2	L2	No
Order 08	Accepted	Vehicle 06	L2	L2	No
Order 09	Accepted	Vehicle 04	L2	L2	No
Order 10	Accepted	Vehicle 02	L1	L1	No

2. Fleet Relocation Schedule

Veh.	Task Type	From	To	Departs	Arrives	Task Description
V01	Standby	S1	S1	01/03 12:00	01/03 12:00	Fulfill initial order (Order 04)
V02	Relocation	S2	S4	01/01 02:00	01/01 06:00	Fulfill initial order (Order 10)
V03	Relocation	S3	S5	01/02 00:00	01/02 03:30	Fulfill initial order (Order 03)
V04	Relocation	S5	S3	01/01 16:00	01/01 19:30	Fulfill initial order (Order 09)
V04	Relocation	S4	S3	01/04 15:30	01/04 18:30	Relocate for Order 06 after completing Order 09
V05	Relocation	S4	S2	01/01 23:30	01/02 03:30	Fulfill initial order (Order 07)
V06	Standby	S6	S6	01/01 00:00	01/01 00:00	Fulfill initial order (Order 05)
V06	Consecutive Service	S6	S6	01/04 20:00	01/04 20:00	Proceed to Order 08 after completing Order 05

* Note: Veh. = Vehicle, S = Station.